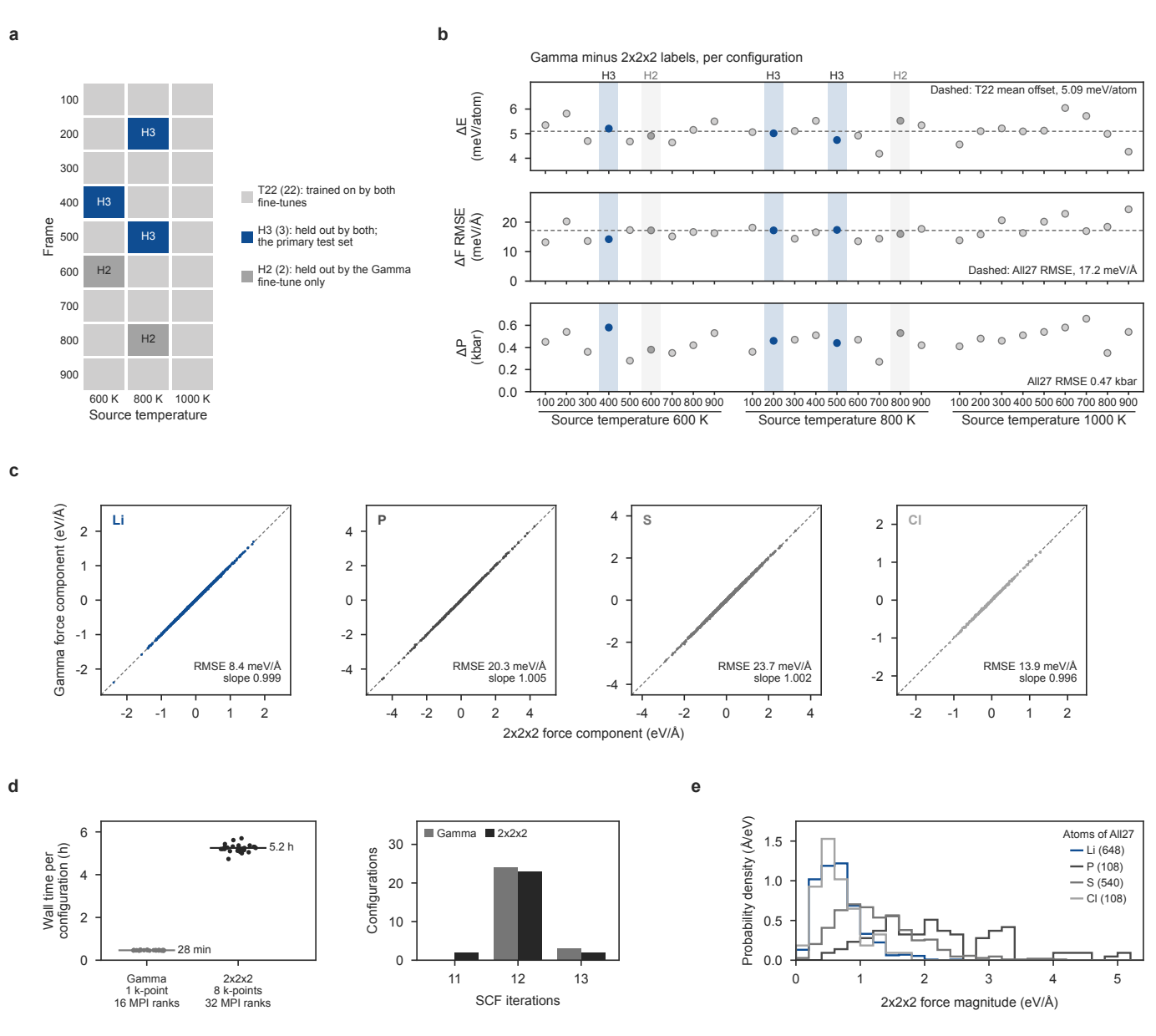

a
Frame
100
200
300
400
500
600
700
800
900
H3
H3
H3
H2
H2
600 K
800 K
1000 K
Source temperature
T22 (22): trained on by both fine-tunes
H3 (3): held out by both; the primary test set
H2 (2): held out by the Gamma fine-tune only
b
Gamma minus 2x2x2 labels, per configuration
H3
H2
H3
H3
H2
Dashed: T22 mean offset, 5.09 meV/atom
ΔE (meV/atom)
ΔF RMSE (meV/Å)
Dashed: All27 RMSE, 17.2 meV/Å
ΔP (kbar)
All27 RMSE 0.47 kbar
100 200 300 400 500 600 700 800 900
Source temperature 600 K
Source temperature 800 K
Source temperature 1000 K
c
Li
RMSE 8.4 meV/Å
slope 0.999
P
RMSE 20.3 meV/Å
slope 1.005
S
RMSE 23.7 meV/Å
slope 1.002
Cl
RMSE 13.9 meV/Å
slope 0.996
Gamma force component (eV/Å)
2x2x2 force component (eV/Å)
d
Wall time per configuration (h)
5.2 h
28 min
Gamma
1 k-point
16 MPI ranks
2x2x2
8 k-points
32 MPI ranks
Configurations
Gamma
2x2x2
SCF iterations
e
Probability density (Å/eV)
Atoms of All27
Li (648)
P (108)
S (540)
Cl (108)
2x2x2 force magnitude (eV/Å)